

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

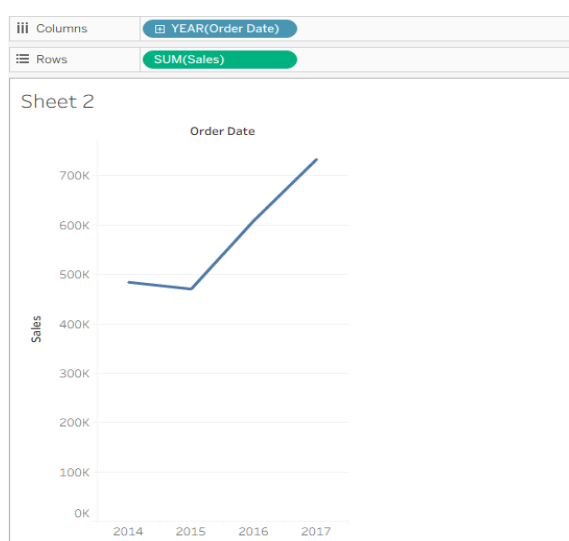
Practical - V

Roll No.: T032	Name: KHUSHI
Class: TYIT	Batch: 2
Date of Assignment: 20-8-26	Date/Time of Submission: 20-8-26

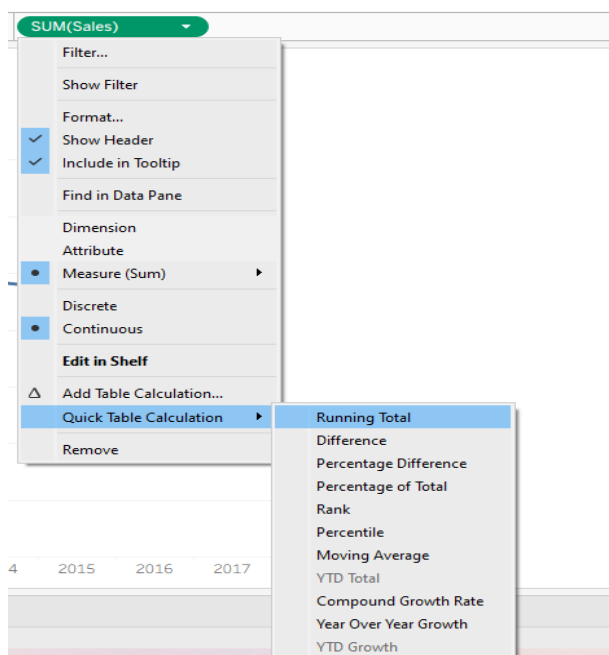
Analyzing Data using Table Calculations in Tableau

A] Apply quick table calculations (running total, percent of total)

- Open Tableau → Connect → **Text File** → select `superstore.csv`.
- Drag **Order Date** → Columns.
- Drag **Sales** → Rows.

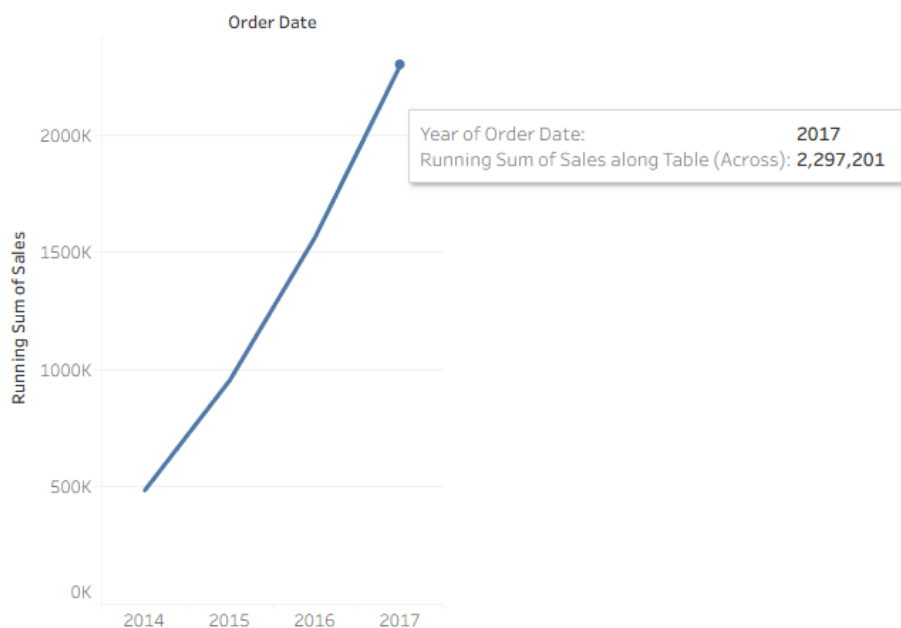


- Click the drop-down on `SUM(Sales)`.
- Select **Quick Table Calculation** → **Running Total**.



Quick Table Calculations

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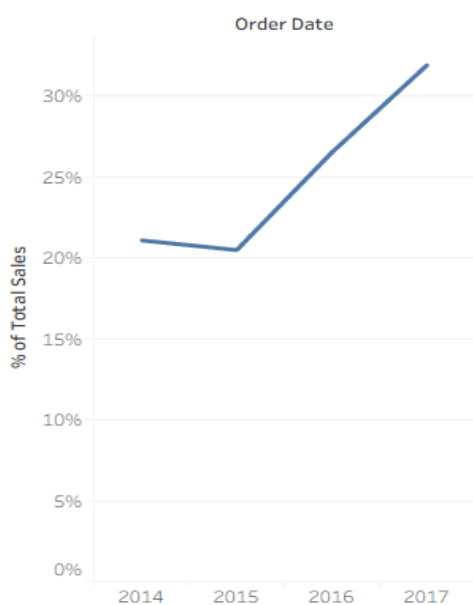


2. Percent of Total

- Keep **Order Date** → Columns.
- Keep **Sales** → Rows.
- Select `SUM(Sales)` → **Quick Table Calculation** → **Percent of Total**.

Quick Table Calculations

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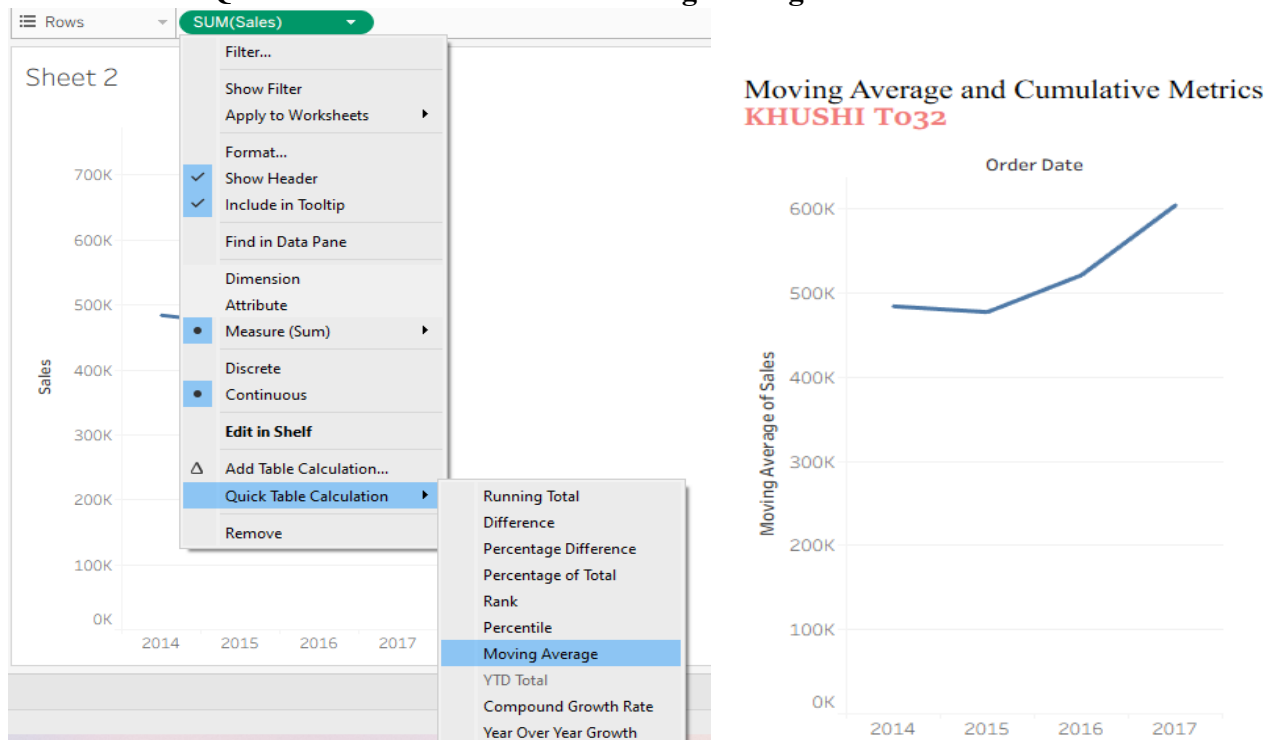


b) Create moving averages and cumulative metrics

Moving Average :

- Drag **Order Date** → Columns.
- Drag **Sales** → Rows.
- Select `SUM(Sales)`.

- Choose **Quick Table Calculation** → **Moving Average**.



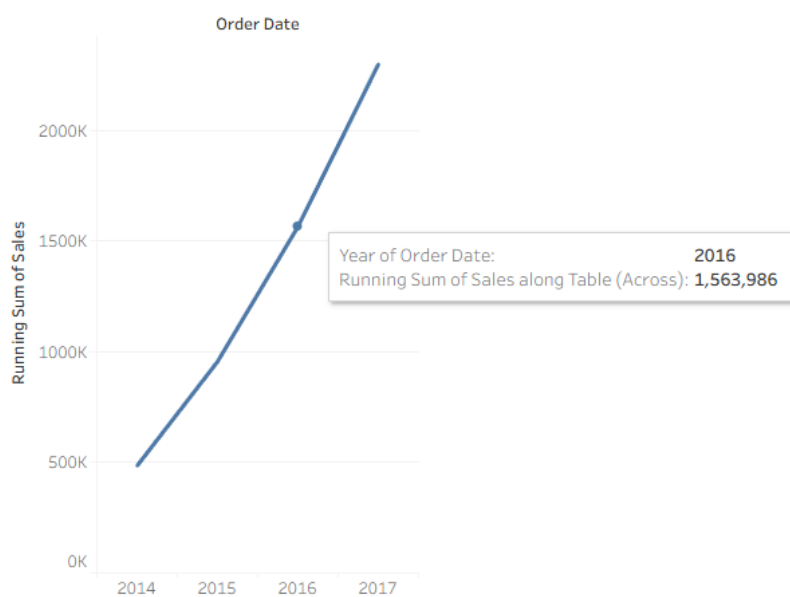
Cumulative Sales :

Use:

Sales → **Quick Table Calculation** → **Running Total**

Moving Average and Cumulative Metrics

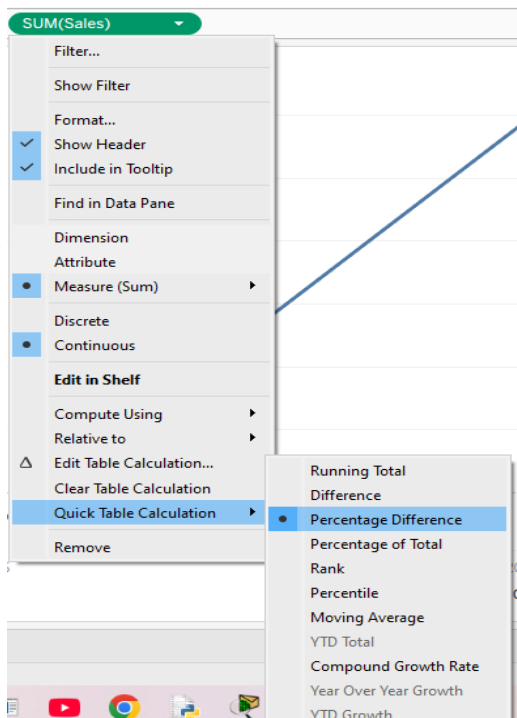
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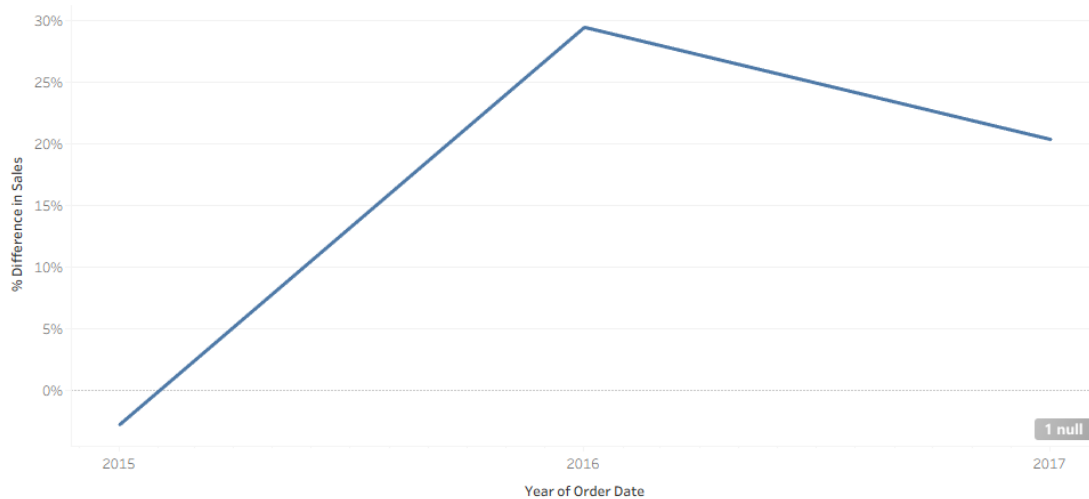
This gives cumulative sales over time.

c) Perform year-over-year growth analysis

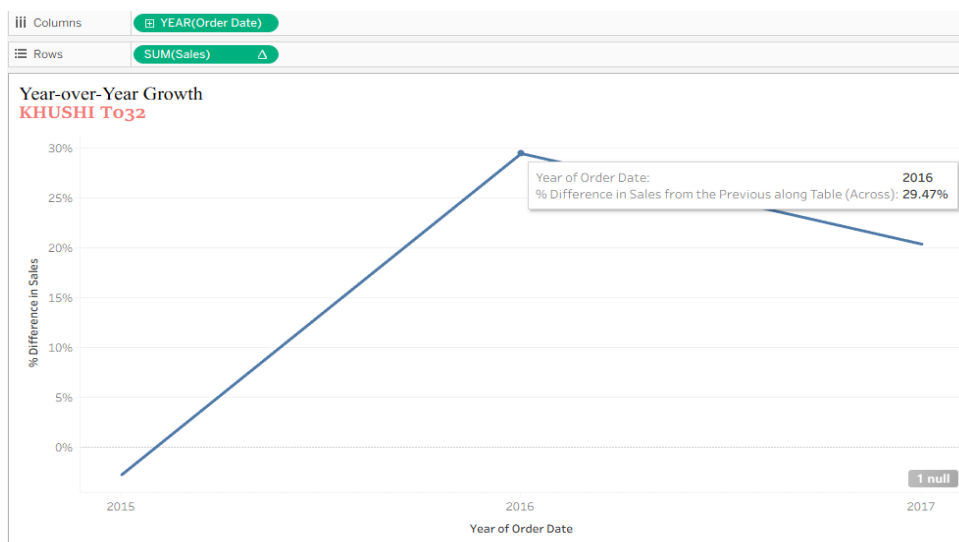
- Put **Order Date** → Columns.
- Change it to **YEAR(Order Date)**.
- Put **Sales** → Rows.
- Right-click **SUM(Sales)** → **Quick Table Calculation** → **Percentage Difference**.



Year-over-Year Growth
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- Right-click SUM(Sales) again → **Edit Table Calculation.**
- Set **Relative to** → **Previous.**



d) Use ranking functions (Top N, Bottom N)

- Drag **Product Name** → Rows.
- Drag **Sales** → Columns.
- Sort descending.
- Select **Product Name** → **Filter**.

Product Name

Edit Filter...

✓ Show Filter

Show Highlighter

Apply to Worksheets

Rows

Product Name

Edit Filter...

✓ Show Filter

Show Highlighter

Apply to Worksheets

Sort...

Sheet 4

Product Name

Zipper Ring Binder P

Zebra ZM400 Thermo

Zebra Zazzle Fluores

Product Name

Product Name

Filters

Product Name

Edit Filter...

✓ Show Filter

Show Highlighter

Sheet 4

Prod

ber Ring l

ra ZM400

ra Zazzle

A Filter [Product Name] window will open.

Filters section → blue “Product Name” pill → click the tiny ▼/drop-down on the RIGHT side of that blue pill.

Then select:

Edit Filter... → Top tab → By field → Top 10 → SUM → Sales → OK

Filter [Product Name]

General Wildcard Condition Top

☐ None

☒ By field:

Top 10 by

Sales Sum

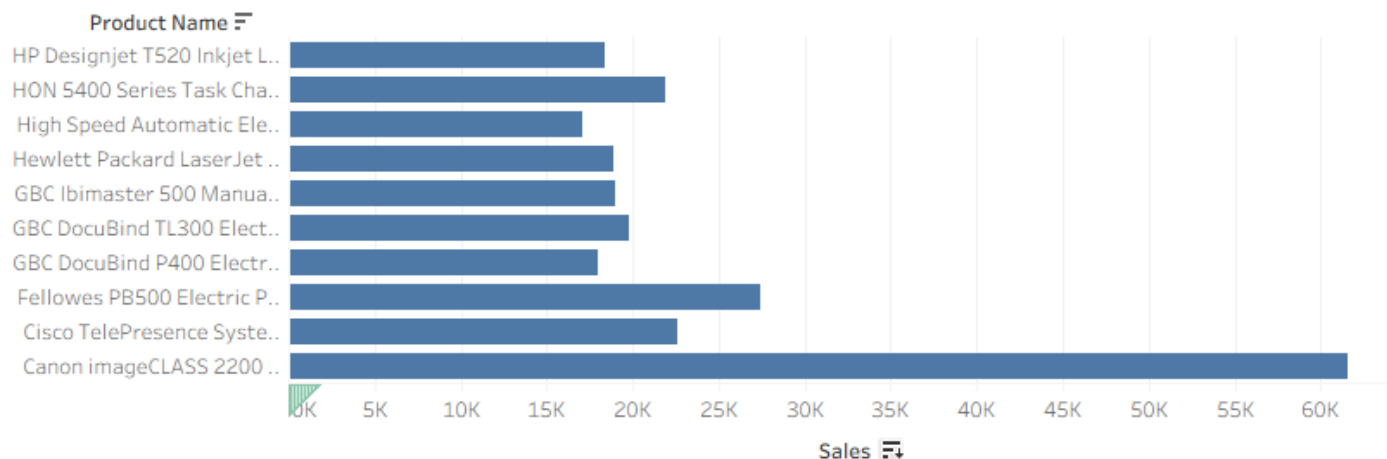
☐ By formula:

Top 10 by

Reset OK Cancel Apply

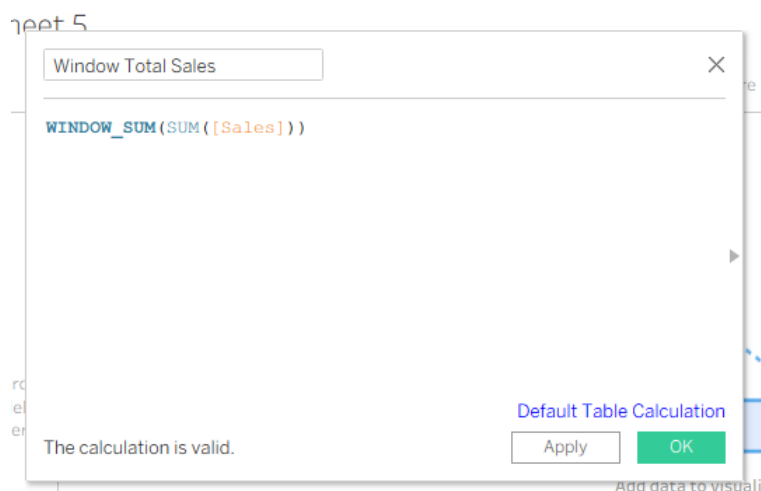
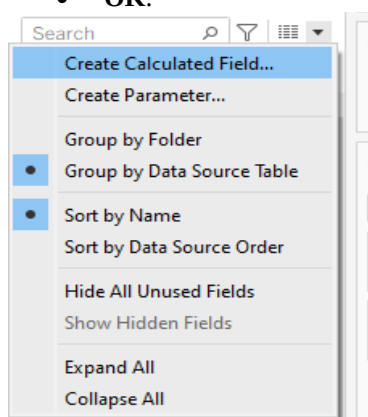
Year-over-Year Growth

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e) Implement window functions (WINDOW_SUM, WINDOW_AVG)

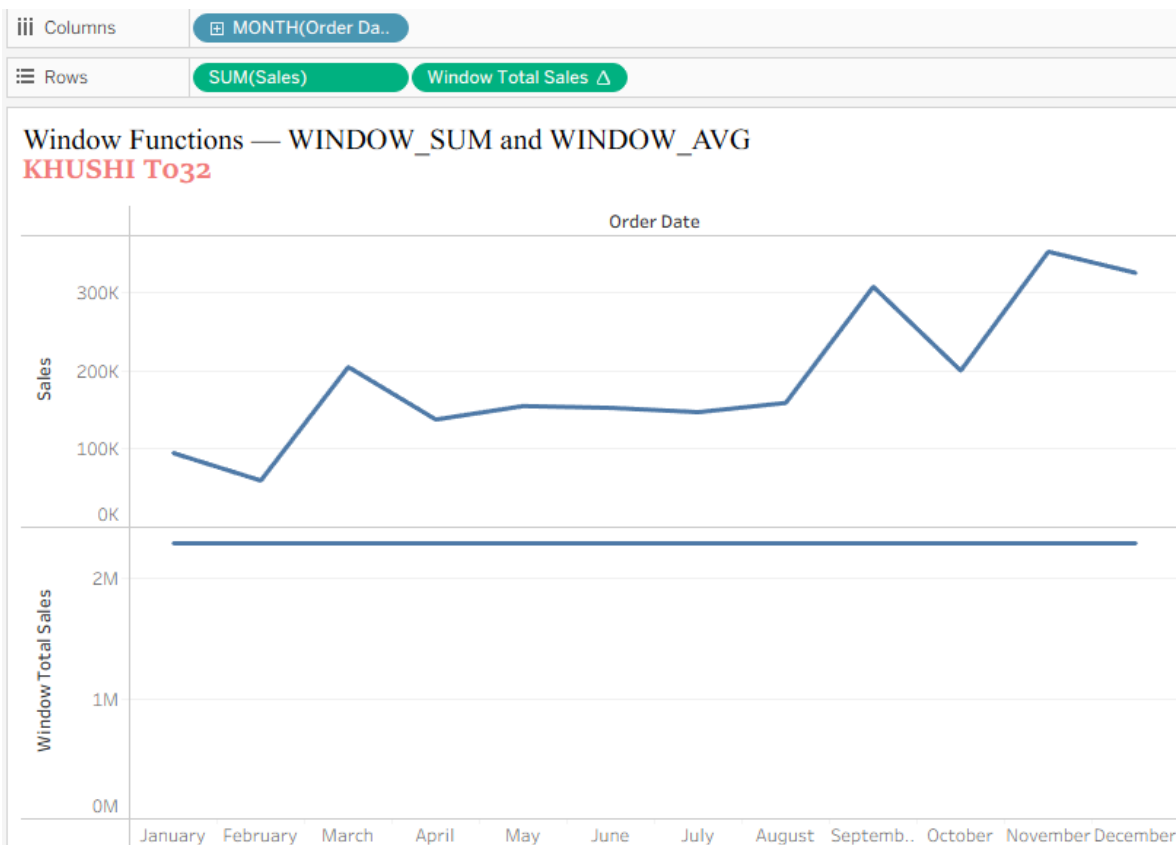
- Create a calculated field:
- **WINDOW_SUM**
- Go to **Analysis** → **Create Calculated Field**.
- Name:
- Window Total Sales
- Formula: `WINDOW_SUM(SUM([Sales]))`
- **OK**.



Order Date → Columns

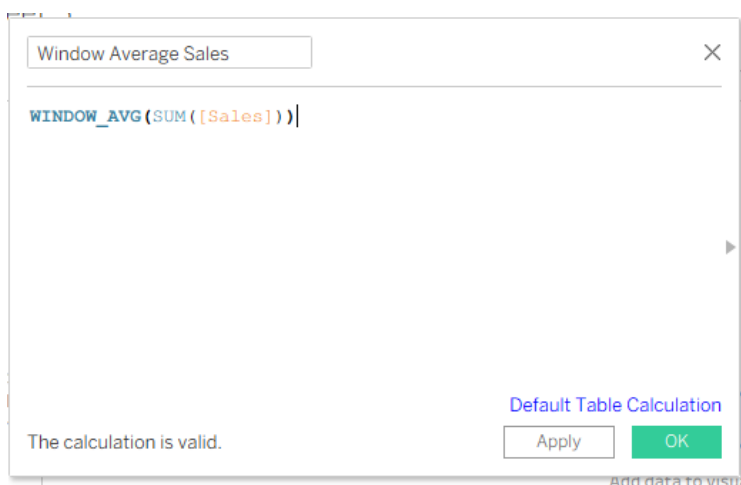
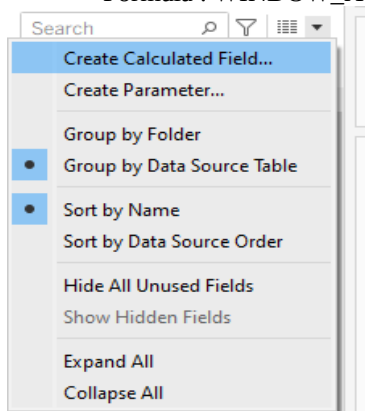
Sales → Rows

Drag **Window Total Sales** → **Rows**.



WINDOW_AVG :

- Create another calculated field:
- Window Average Sales
- Formula : WINDOW_AVG(SUM([Sales]))

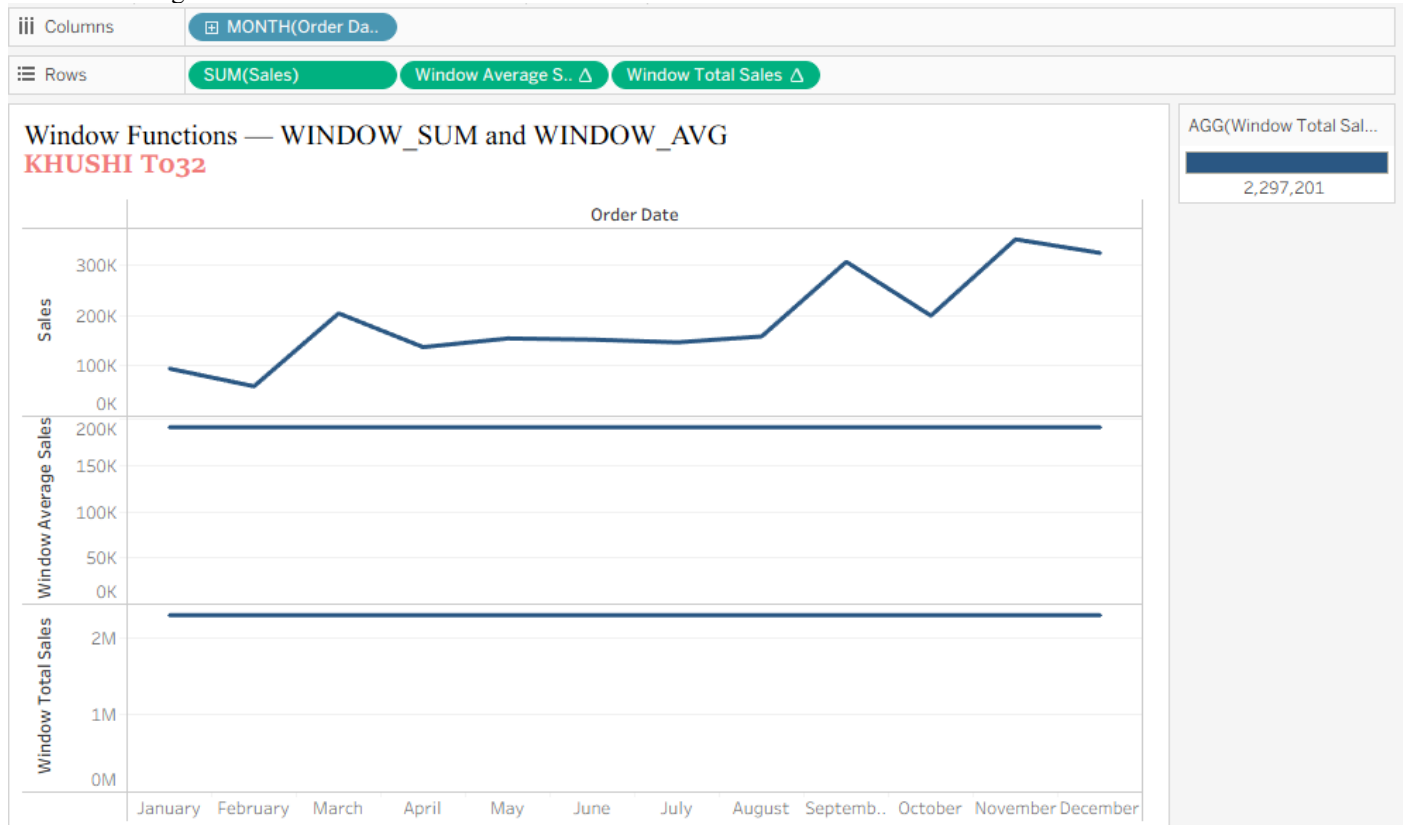


Order Date → Columns

Sales → Rows

Drag **Window Total Sales** → Rows

Window Average Sales → Rows



f) Customize addressing and partitioning

1] Table Across

You already have **Table (across)** selected. ✓

- Click **Table (across)**
- Observe the result in your table

Columns
YEAR(Order Date)

Rows
Category

Customize Addressing and Partitioning
KHUSHI T032

	Order Date			
Category	2014	2015	2016	2017
Furniture	157,193	327,711	526,613	742,000
Office Supplies	151,776	289,010	472,950	719,047
Technology	175,278	338,059	564,423	836,154

Table Calculation
Running Sum of Sales

Calculation Type

Running Total

Sum

Compute Using

Table (across)

Table (down)

Table (across then down)

Table (down then across)

Cell

Specific Dimensions

☒ Year of Order Date

2] Table Down

Now click:

🔑 Table (down)

Observe the result.

Columns

YEAR(Order Date)

Rows

Category

Customize Addressing and Partitioning

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		Order Date			
Category	2014	2015	2016	2017	
Furniture	157,193	170,518	198,901	215,387	
Office Supplies	308,969	307,752	382,841	461,484	
Technology	484,247	470,533	609,206	733,215	

Table Calculation

Running Sum of Sales

Calculation Type

Running Total

Sum

Compute Using

Table (across)

Table (down)

Table (across then down)

Table (down then across)

Cell

Specific Dimensions

☒ Category

3] Pane Across

For this one, click:

🔑 Specific Dimensions

Then you can work with the dimensions shown below:

- ☒ Year of Order Date
- ☐ Category

Columns

YEAR(Order Date)

Rows

Category

Customize Addressing and Partitioning

KHUSHI T032

		Order Date			
Category	2014	2015	2016	2017	
Furniture	157,193	327,711	526,613	742,000	
Office Supplies	151,776	289,010	472,950	719,047	
Technology	175,278	338,059	564,423	836,154	

Table Calculation

Running Sum of Sales

Calculation Type

Running Total

Sum

Compute Using

Table (across)

Table (down)

Table (across then down)

Table (down then across)

Cell

Specific Dimensions

☐ Category
☒ Year of Order Date

Restarting every

Sort order

Specific Dimensions

☐ Add secondary calculation
☒ Show calculation assistance

4] Pane Down

Similarly, use:

Specific Dimensions

and adjust the selected dimensions so the calculation works **down the pane**.

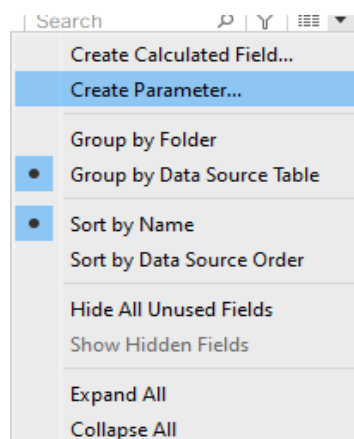
Customize Addressing and Partitioning

KHUSHI T032

Category	Order Date			
	2014	2015	2016	2017
Furniture	157,193	327,711	526,613	742,000
Office Supplies	151,776	289,010	472,950	719,047
Technology	175,278	338,059	564,423	836,154

g) Create dynamic titles and labels using table calculations

- **Worksheet → Show Title**
- Double-click the **chart title**.
- Delete the old title.
- Type **Sales Trend for**
- Click **Insert → Parameters → Select Category**.
- Click **OK**.
- Change **Select Category** to see the title change automatically.



Create Parameter

×

Name

Select Category

Properties

Data type

String

Display format

Furniture

Current value

Furniture

Value when workbook opens

Current value

Allowable values

☐ All
☒ List
☐ Range

Value	Display As
Furniture	Furniture
office supplies	office supplies
Technology	Technology
Click to add	

☒ Fixed
☐ When workbook opens

Add values from ▾

Remove Selected

Cancel

OK



Selected Sales

×

```
IF [Category] = [Select Category] THEN [Sales]
END
```



The calculation is valid.

Apply

OK

